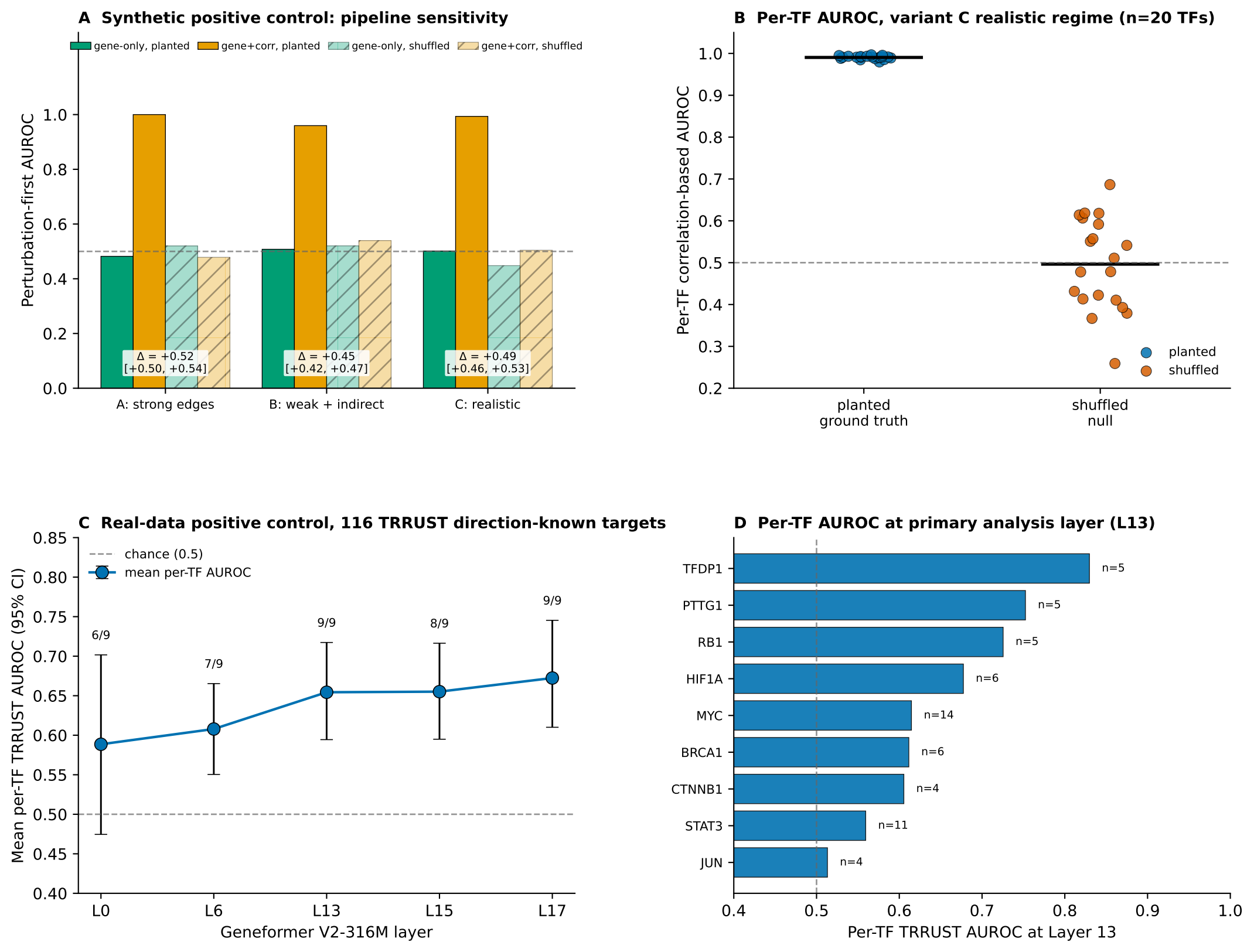
A Synthetic positive control: pipeline sensitivity
gene-only, planted
gene+corr, planted
gene-only, shuffled
gene+corr, shuffled
Perturbation-first AUROC
Δ = +0.52
[+0.50, +0.54]
Δ = +0.45
[+0.42, +0.47]
Δ = +0.49
[+0.46, +0.53]
A: strong edges
B: weak + indirect
C: realistic
B Per-TF AUROC, variant C realistic regime (n=20 TFs)
Per-TF correlation-based AUROC
planted
shuffled
planted
ground truth
shuffled
null
C Real-data positive control, 116 TRRUST direction-known targets
chance (0.5)
mean per-TF AUROC
Mean per-TF TRRUST AUROC (95% CI)
6/9
7/9
9/9
8/9
9/9
L0
L6
L13
L15
L17
Geneformer V2-316M layer
D Per-TF AUROC at primary analysis layer (L13)
TFDP1
PTTG1
RB1
HIF1A
MYC
BRCA1
CTNNB1
STAT3
JUN
n=5
n=5
n=5
n=6
n=14
n=6
n=4
n=11
n=4
Per-TF TRRUST AUROC at Layer 13